PROBLEM 4
BILL OF MATERIALS FOR AN RC AIRCRAFT

Technical proposal, budget and interview defence guide

Brief: 1–1.2 m wingspan; conventional or high wing; one electric motor; wireless control; lightweight, repairable structure; ground or hand launch.

Important: This is a planning BOM, not a purchasing list. Prices are broad illustrative INR estimates and have not been verified as live supplier quotations. Confirm exact parts, compatibility and current prices before submission.

1. Recommended concept

A high-wing foam-board aircraft with a conventional tail is a practical starting point because it is inexpensive, quick to repair and suitable for early prototyping. Reinforce the wing spar and high-load joints with wood or carbon strips. The motor, propeller, ESC and battery must be treated as one matched power system and checked against manufacturer data.

2. Bill of materials and alternatives

Item / qty	Option A	Option B	Key checks	Indicative ■
Airframe, 1	5 mm foam board	Light EPP foam kit	1–1.2 m span; repairability	600–1,800
Wing reinforcement, 1 set	Pine/basswood strip	Carbon strip/tube	Stiffness, bonding and mass	150–700
Brushless motor, 1	2212-class, ~1000–1400 KV candidate	Higher-quality matched motor	Use maker thrust/current/prop chart; KV alone is insufficient	500–1,400
ESC, 1	30 A ESC with BEC	40 A ESC with suitable BEC	Continuous current, cell count and BEC compatibility	500–1,200
Propeller, 2–3	Maker-approved 8–9 in candidate	Alternative approved size/pitch	Must match motor, voltage and current limits	100–350
Battery, 1–2	3S LiPo, 1300–2200 mAh	Higher-quality compatible 3S pack	Capacity, discharge rating, connector, mass and voltage sag	900–2,000 each
Charger, 1	3S LiPo balance charger	Smart balance charger with storage mode	Correct chemistry/cell count; safe charging	900–2,500
Radio set, 1	Entry-level matched 6-channel set	Telemetry-capable matched set	Protocol compatibility, failsafe, range and channels	3,000–8,000
Servos, 4	9 g analog micro servos	Metal-gear micro servos where needed	Torque, voltage, travel, mounting	600–1,600 total
Horns/linkages, 1 set	Plastic horns + wire pushrods	Stiffer pushrods/adjustable clevises	No binding; secure retainers	200–600
Landing gear, optional	Wire gear + small wheels	Light commercial gear	Ground clearance and secure mounting	200–700
Connectors/wiring, 1 set	Compatible connector and wire	Higher-quality connectors/wire	Polarity, current rating, soldering and strain relief	250–700
Adhesives/tape, 1 set	Foam-safe glue and tape	Epoxy for suitable joints	Foam compatibility and cure time	250–700
Tools/safety allowance	Basic hand tools/prop wrench	Add wattmeter and thrust scale	Secure stand, eye protection, LiPo safety	500–2,500

Retailer/source column for final submission: cite the exact manufacturer datasheet for the motor, ESC, propeller, battery, radio, servos and charger, plus current Indian retailer listing(s). No exact supplier links are asserted here because the brief did not specify exact selected part numbers.

3. Budget configurations

Configuration	Estimated total	Advantages	Trade-offs
Budget build	■8,000–■13,000	Affordable, repairable structure and basic electronics.	Less telemetry; depends on ownership of radio/charger/tools.
Performance-oriented	■13,000–■22,000	Potentially lighter/stiffer build, better radio features and test tools.	Higher cost; correct design/testing still determines performance.

Totals are planning estimates, not supplier quotes. Recalculate using selected parts and include transmitter, receiver, charger, tools, delivery and spare batteries if they must be purchased. Recommended competition choice: reliable budget-to-midrange build; prioritize compatible radio, proven powertrain, reinforced wing and safe test equipment over cosmetics.

4. Component selection rationale

Motor and propeller: KV is approximate no-load RPM per volt; it does not specify thrust or efficiency by itself. Use the motor maker's propeller/current/thrust data at the intended voltage. Confirm with a safely restrained static test using a wattmeter and thrust scale. Do not exceed component limits.

ESC: Select continuous current rating above expected/measured current with sensible margin (about 20–30% can be a preliminary allowance), and check battery cell count, cooling and BEC voltage/current for servos. Margin is not a substitute for compatibility.

Battery: 3S LiPo has nominal voltage 11.1 V (3 × 3.7 V) and full-charge voltage about 12.6 V. Capacity affects endurance and mass. Discharge rating must support current demand; check voltage sag and connector compatibility. Use a suitable balance charger and safe storage/handling.

Servos: Four assumed: two ailerons, one elevator and one rudder. Servo count can change with the design. Verify torque, voltage, travel and linkage geometry.

Structure: Foam board/EPP is inexpensive and repairable; wood spar adds stiffness; carbon may be lighter/stiffer but costs more and requires good joints. Strength depends on the complete structure and bonding.

Compatibility checks: Motor mount/shaft and prop adapter; propeller diameter/pitch and rotation; battery connector and polarity; ESC current/voltage/BEC; servo voltage/connectors; transmitter-receiver protocol; channel count and failsafe; component packaging and CG.

5. Takeoff mass and propulsion sufficiency (bonus)

Illustrative mass budget: airframe/reinforcement 350–450 g; motor/ESC/propeller 100–160 g; battery 120–200 g; receiver and four servos 70–110 g; wiring/linkages/adhesives/landing gear 100–180 g. This gives roughly 740–1,100 g before design-specific additions. Use 0.9–1.2 kg as a preliminary planning target and weigh the finished aircraft.

For 1.0 kg, weight $W = m\,g = 1.0 \times 9.81 = 9.81$ N. Example: measured static thrust 6 N gives $T/W = 6/9.81 \approx 0.61$. A static thrust-to-weight ratio around 0.5–0.8 can be a preliminary target for a simple trainer, not a universal pass/fail rule. Wing loading, drag, prop efficiency, launch method and available airspeed also matter.

Wing loading = mass / wing area (commonly g/dm²). Stall speed depends on wing loading and lift coefficient; the brief provides no wing area or lift coefficient, so a defensible numeric stall speed cannot be calculated from the supplied information.

6. Interview defence questions

What does 3S mean? Three LiPo cells in series; nominal 11.1 V and fully charged about 12.6 V.

Why is KV not enough to choose a motor? KV gives approximate no-load RPM per volt, not thrust/current for a particular propeller.

How is thrust-to-weight ratio calculated? Thrust force divided by aircraft weight force, in consistent units.

Why is CG important? Incorrect centre of gravity can make the aircraft unstable or prevent safe rotation/climb; follow the design CG range.

Why four servos? Two ailerons, elevator and rudder provide conventional roll, pitch and yaw control.

Which figures are estimates? Prices, mass budget, component classes and performance targets until confirmed by datasheets and measurements.

Safety and verification

The uploaded brief does not include exact part numbers or current supplier links. Before final submission, replace broad estimates with chosen parts, current source links and their datasheets. Do not claim testing was performed unless it was. LiPo batteries and rotating propellers are hazardous: use correct charging/storage practice, remove the propeller during radio setup, secure the model for powered tests, and test flight only in a suitable legal area with experienced supervision.